2026

COMPUTER PROGRAMING

ASSIGNMENT NO # 1

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REG NO: BF25NWELE0710

SECTION: ELECTRICAL POWER

SUBMITTED TO: SIR RIZWAN

Write a single C++ statement to accomplish each of the following:

i. Declare the variables voltage, current, and resistance to be of type double.

```
double voltage, current, resistance;
// Create three decimal-number variables to store voltage, current, and
resistance
```

ii. Display the message "Welcome to Object Oriented Programming in C++!" on the screen.

```
cout << "Welcome to Object Oriented Programming in C++!";
// Show a welcome message on the screen for the user
```

iii. Read a value from the user and store it in an integer variable named age.

```
cin >> age;
// Take age input from the keyboard and store it inside variable age
```

iv. If the variable temperature is greater than 100, print "Overheating detected!".

```
if (temperature > 100) cout << "Overheating detected!";
// Check temperature; if it crosses 100, warn about overheating
```

v. Compute the area of a circle with radius r and store it in area ($\pi = 3.14159$).

```
area = 3.14159 * r * r;
// Calculate circle area using πr² and save the result in area
```

vi. Print the values of x, y, and z separated by a comma.

```
cout << x << "," << y << "," << z;
// Print x, y, and z neatly separated by commas
```

vii. Assign the sum of voltage and current to total.

```
total = voltage + current;
// Add voltage and current, then store the result in total
```

viii. Display “Error: Division by zero” only if denominator equals zero.

```
if (denominator == 0) cout << "Error: Division by zero";
// Show error message only when denominator becomes zero
```

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ix. Increment the variable count by 1 and then print its new value.

```
cout << ++count;
// Increase count by 1 first, then print the updated value
```

---

x. Read three floating-point numbers from the keyboard into a, b, and c.

```
cin >> a >> b >> c;
// Take three decimal numbers from user and store them in a, b, and c
```

---

xi. Total resistance in series (take values from user).

```
cin >> R1 >> R2 >> R3, cout << "Total Resistance (Series) = " <<
(R1 + R2 + R3);
// Take three resistor values and add them because series resistances simply
add
```

---

xii. Total resistance in parallel (take values from user).

```
cin >> R1 >> R2 >> R3, cout << "Total Resistance (Parallel) = "
<< 1 / ((1/R1) + (1/R2) + (1/R3));
// Take three resistances and apply parallel resistance formula
```

---

xiii. Voltage across R2 using Voltage Division Rule.

(Chosen values: R1 = 10Ω, R2 = 20Ω, V = 12V)

```
cout << "Voltage across R2 = " << 12 * (20.0 / (10 + 20));
// Use voltage division rule to calculate voltage across R2
```

---

xiv. Current through R2 using Current Division Rule.

(Chosen values: R1 = 10Ω, R2 = 20Ω, V = 12V)

```
cout << "Current through R2 = " << (12.0 / (10 + 20)) * (10.0 /
(10 + 20));
```

```
// Find total current then apply current division rule to get current through R2
```

---

xv. Calculate Semester GPA.

```
cin >> credit >> grade, totalCredits += credit, totalPoints += credit * grade;
// Enter credit hours and grade points, then update totals

cout << "Semester GPA = " << (totalPoints / totalCredits);
// Divide total grade points by total credit hours to calculate GPA
```
